

AI PROJECT · SPRING 2026

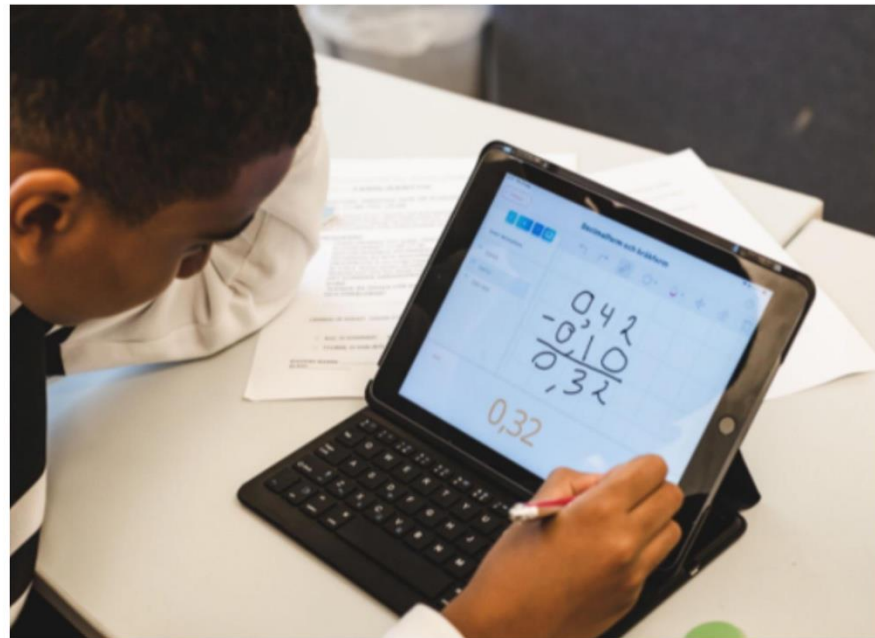
# Draw & Solve

Handwritten Math Expression Recognition

| Name                         | Student ID | Task                            |
|------------------------------|------------|---------------------------------|
| Matti Hannes Vincent Lehmann | T14H06309  | Presentation Slides             |
| David Axel Spickenheuer      | T14H06329  | Data Preparation & Augmentation |
| Jonas Schenke                | T14H06307  | CNN Training & Evaluation       |
| Philip Nguyen                | 91499153X  | Presentation Slides             |
| Staniya Thomas               | T14H06310  | Symbol Recognition Pipeline     |
| Abhirup Sain                 | T14H06318  | Streamlit UI & Integration      |

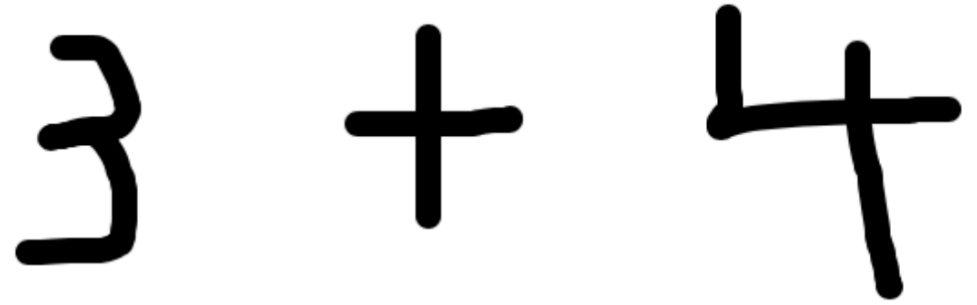
- **Handwritten maths** on tablets and touch screens is increasingly common (online exams, digital whiteboards, teaching apps)
- **Manual typing** of expressions into a calculator or CAS is slow and error-prone
- **Goal:** draw an expression once and obtain an instant, reliable numerical result

**Why now?** Tablet adoption in education has grown rapidly — a seamless, pen-first math solver removes a major friction point for students and teachers.



- Handwriting varies widely per writer
- Operators share features with digits (e.g.  $\times$  vs x,  $-$  vs  $\div$ )
- Multi-stroke digits confuse naive segmenters

**Goal:** Draw a math expression on screen  $\rightarrow$  get the result instantly.

A handwritten mathematical expression "3 + 4" is shown inside a light blue rounded rectangle. The digit "3" is formed with a single continuous stroke, starting from the middle, curving down to the left, then up and around to the right. The plus sign "+" is formed with two strokes: a vertical line and a horizontal line. The digit "4" is formed with two strokes: a vertical line and a horizontal line that crosses it.

1. Draw an expression on a browser canvas
2. OpenCV segments individual symbols
3. CNN classifies each symbol (14 classes)
4. Safe AST evaluator computes the result

## Digits

0 · 1 · 2 · 3 · 4 · 5 · 6 · 7 · 8 · 9

## Operators

+ · − · × · ÷

Detected symbols

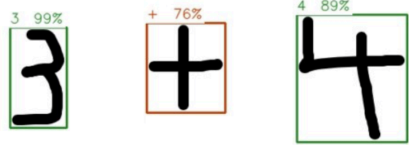
3 + 4

Expression

3+4

Result

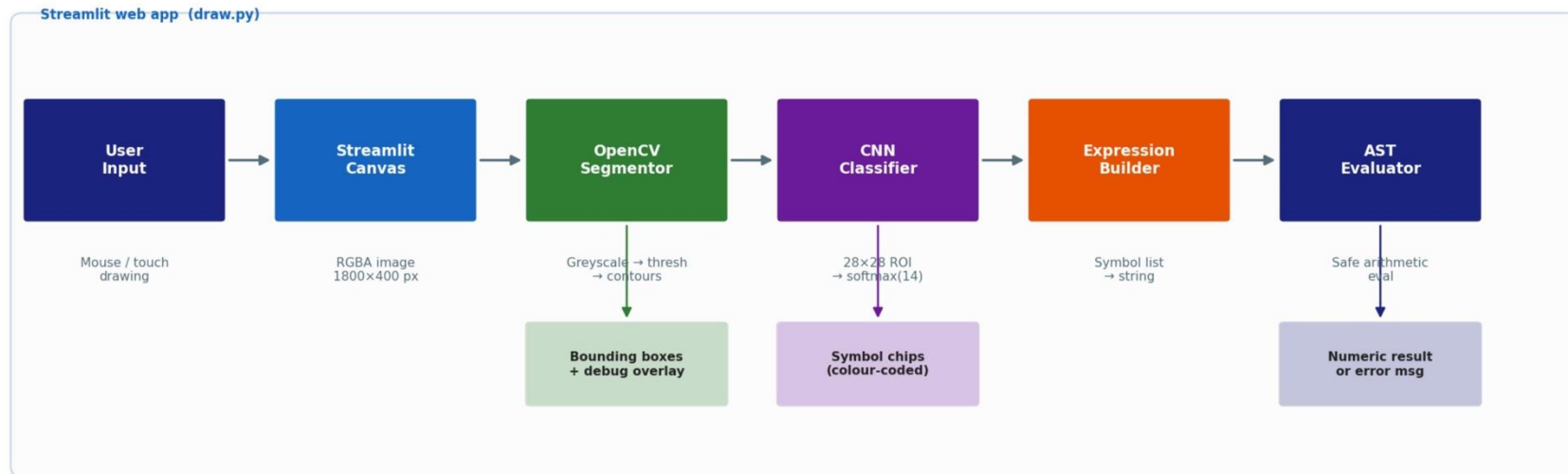
7



Detected symbols (green = digit · teal = operator · orange = uncertain)

The image shows a digital interface for solving a math problem. At the top, it displays 'Detected symbols' as '3 + 4' in colored boxes (green for digits, teal for operators). Below this, the 'Expression' is '3+4' and the 'Result' is '7'. A large white box contains a hand-drawn version of the expression '3+4'. Each symbol in the drawing is enclosed in a colored bounding box with a confidence score: '3' (green, 99%), '+' (orange, 76%), and '4' (green, 89%). A legend at the bottom explains the color coding: green for digits, teal for operators, and orange for uncertain symbols.

## Software Architecture — Draw & Solve



Model: outputs/best\_model.keras · Classes: outputs/class\_names.json

### Frontend

Streamlit · HTML5 Canvas

### Backend

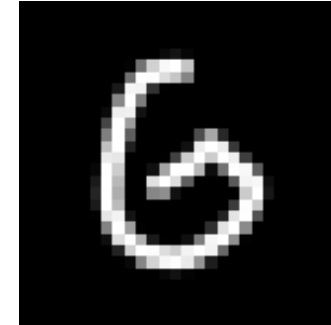
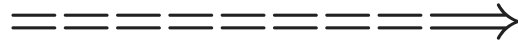
Python · OpenCV · TensorFlow

### Model

Keras · 14-class CNN



How to get from the drawing to the CNN-input?



## First step: Binarisation

- Canvas: white (255), ink  $\approx 0$
- THRESH\_BINARY\_INV: ink  $\rightarrow 255$ , bg  $\rightarrow 0$
- $3 \times 3$  dilation closes gaps between strokes
- findContours detects white regions on black

6 ÷ 3

6 ÷ 3

6 ÷ 3

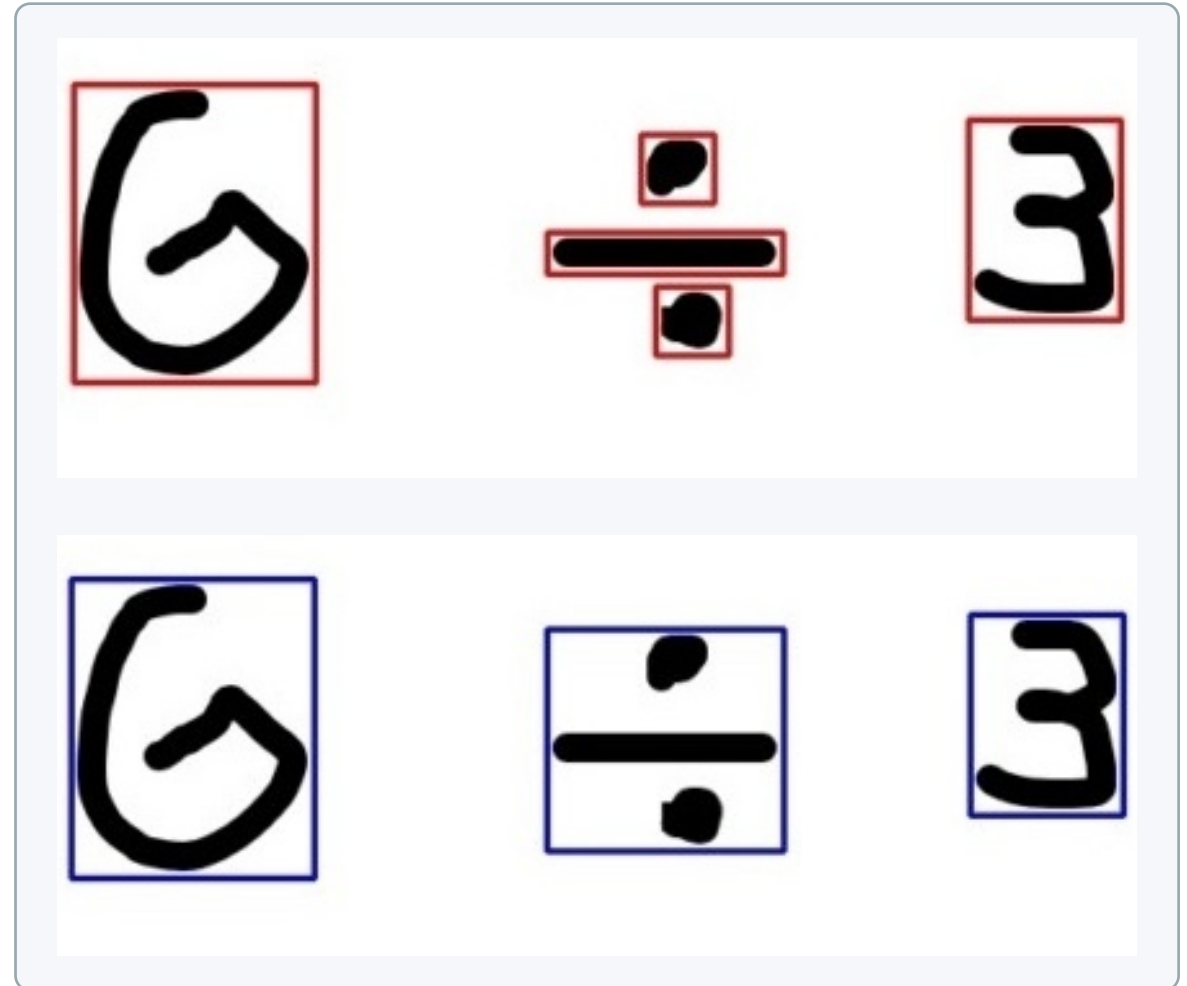
## Bounding boxes

- Outermost contours only (RETR\_EXTERNAL)
- Box area  $< 30 \text{ px}^2$  discarded (noise filter)
- Sort all boxes left  $\rightarrow$  right by x-coordinate

**Problem:** signs and digits can be multi-stroke ( $\div$ , 5, 7)  
 $\rightarrow$  each stroke gets its own box

**Solution:** merge boxes with gap  $\leq 20 \text{ px}$

- Iterate boxes left  $\rightarrow$  right
- If next box starts within 20 px, expand current box





**Final Step:** Conversion of boxed signs to CNN input-format

1. Cropping of Box-content
2. Add padding
3. Downscaling of image to fit in 28 x 28 format
4. centering + additional padding



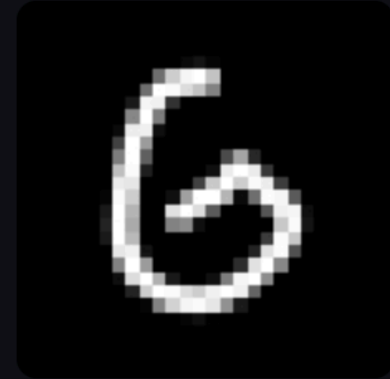
1. Cropped ROI



2. Padding Added



3. Scaled Down



4. Centered in 28x28

# Architecture · Training · Evaluation



**14**

Classes

**177K**

Params

**64192**

Training Samples

## Problem: Imbalance between classes

### Source

Digits 0-9

Operators  $+-\times\div$

### Detail

MNIST (LeCun et al.)

Sagyamthapa (Kaggle)

**14:1 Ratio between MNIST-digits and operators**

## Solution: AUGMENTATION PIPELINE



### Random Rotation

$\pm 8\%$  ( $28.8^\circ$ )

$\pm 8\%$  ( $28.8^\circ$ )



### Random Translation

$\pm 8\%$  H & V

$\pm 8\%$  H & V



### Random Zoom

$\pm 10\%$

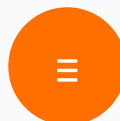
$\pm 10\%$



### Elastic Transform

$\alpha=8$   $\sigma=4$   $p=0.7$

$\alpha=8$   $\sigma=4$   $p=0.7$

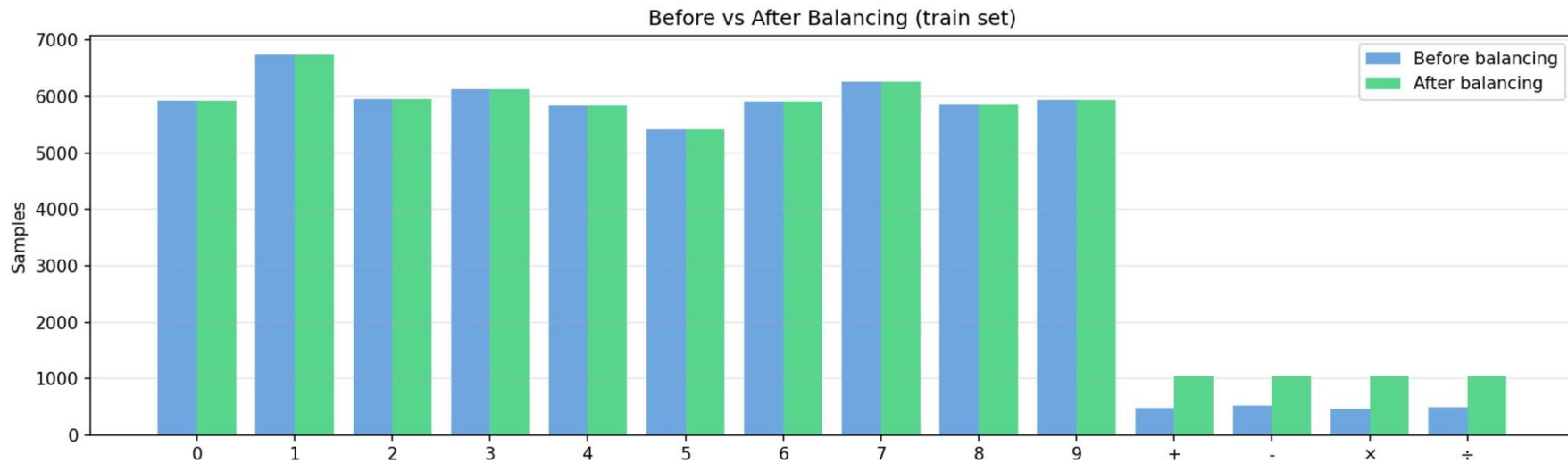


### Pixel Normalize

$\div 255 \rightarrow [0,1]$

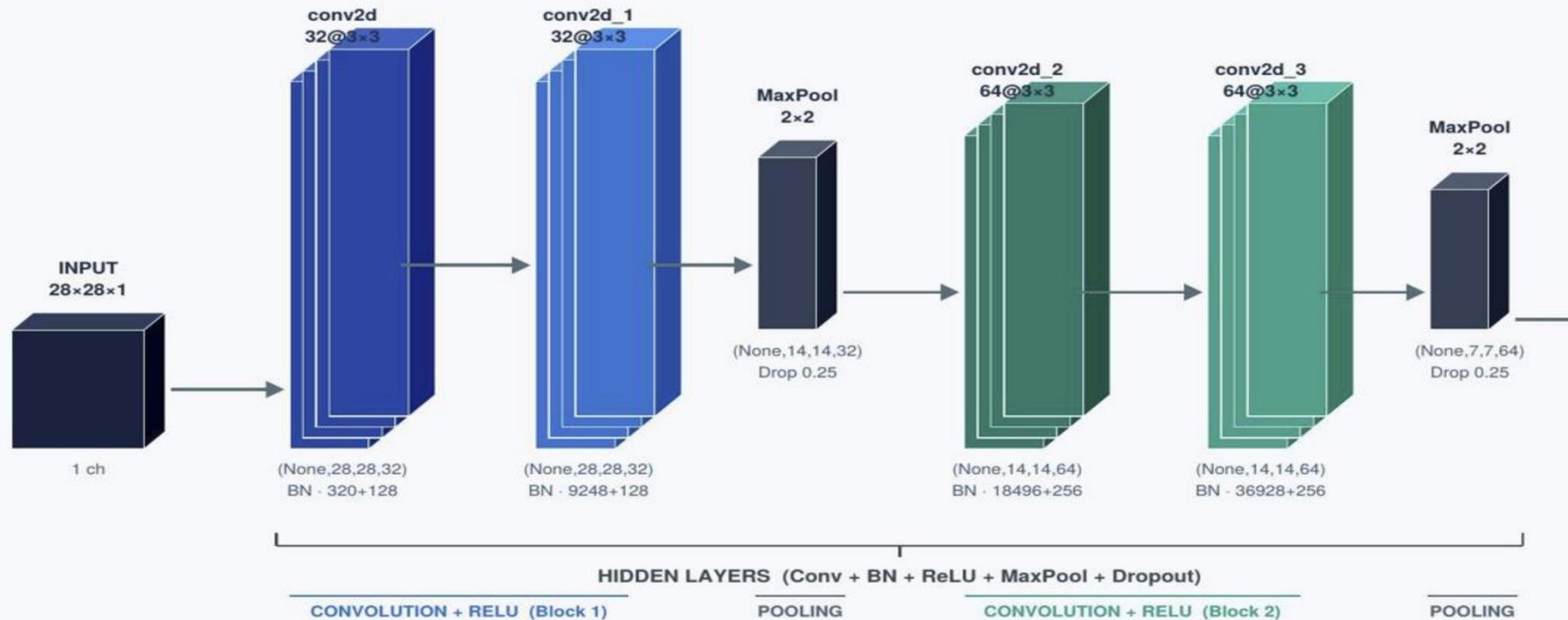
$\div 255 \rightarrow [0,1]$

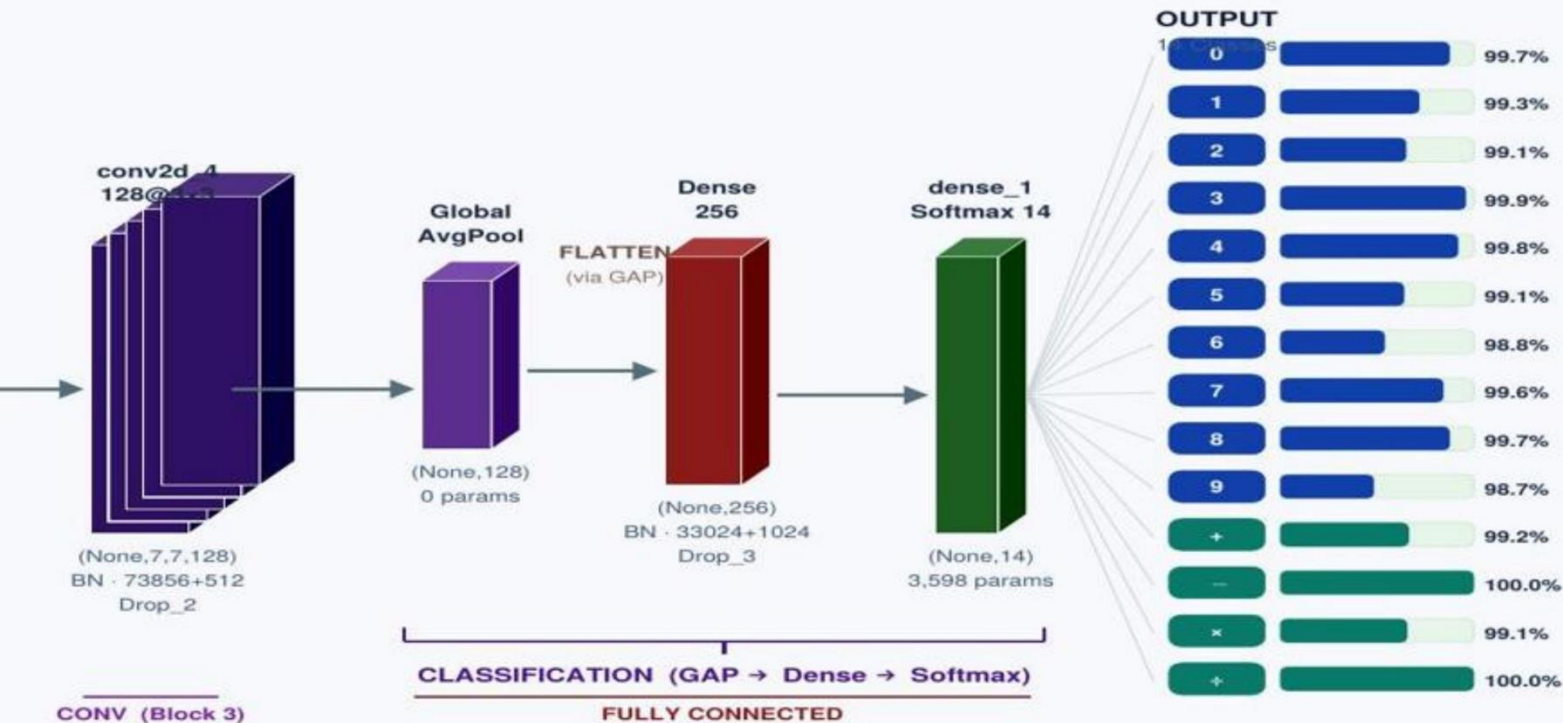
- Digits: 5 000 – 6 700 samples each
- Operators: 1 048 samples each → **6:1 ratio**
- Without fix: model predicts digits almost always



# Convolution Neural Network (CNN) Architecture

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**Batch Normalization** Rescales layer outputs to mean zero and unit variance. Accelerates training by up to 3× and provides mild regularization.

**MaxPooling (2×2)** Downsamples spatial dimensions to reduce computation costs while making detected features invariant to small translational shifts.

**Dropout Regularization** Randomly silences 25–50% of neurons during training. This prevents memorization and forces robust, distributed learning.

**GlobalAveragePooling** Replaces the Flatten layer by averaging feature maps into a compact vector, dramatically reducing model parameter count.

|                   |                                                  |
|-------------------|--------------------------------------------------|
| <b>Optimizer</b>  | AdamW lr=1e-3 wd=1e-4                            |
| <b>Loss</b>       | Categorical Cross-Entropy                        |
| <b>Batch Size</b> | 128                                              |
| <b>Epochs</b>     | 20 (early stop @ 18)                             |
| <b>Best Epoch</b> | Epoch 13                                         |
| <b>Callbacks</b>  | EarlyStopping(patience=5) +<br>ReduceLROnPlateau |

## LR SCHEDULE

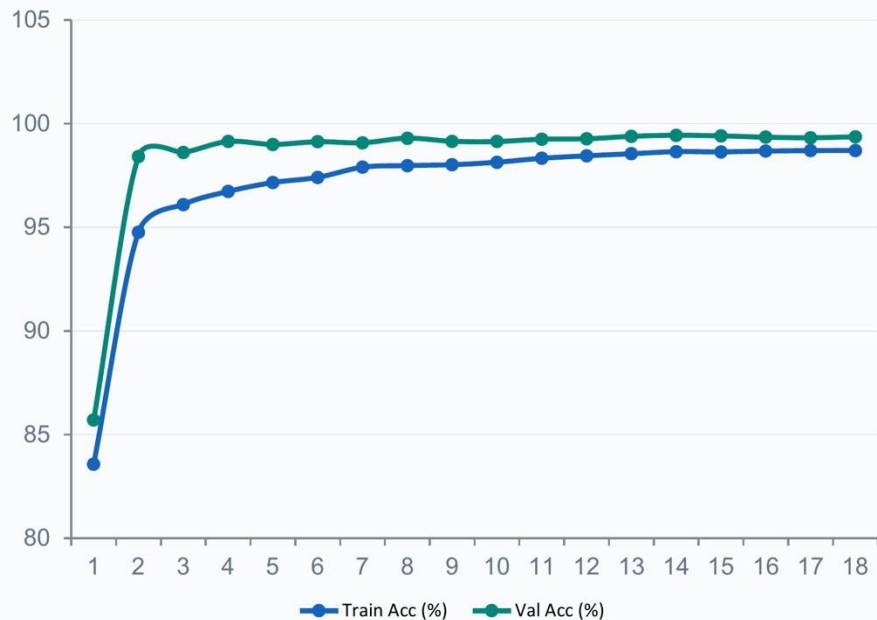




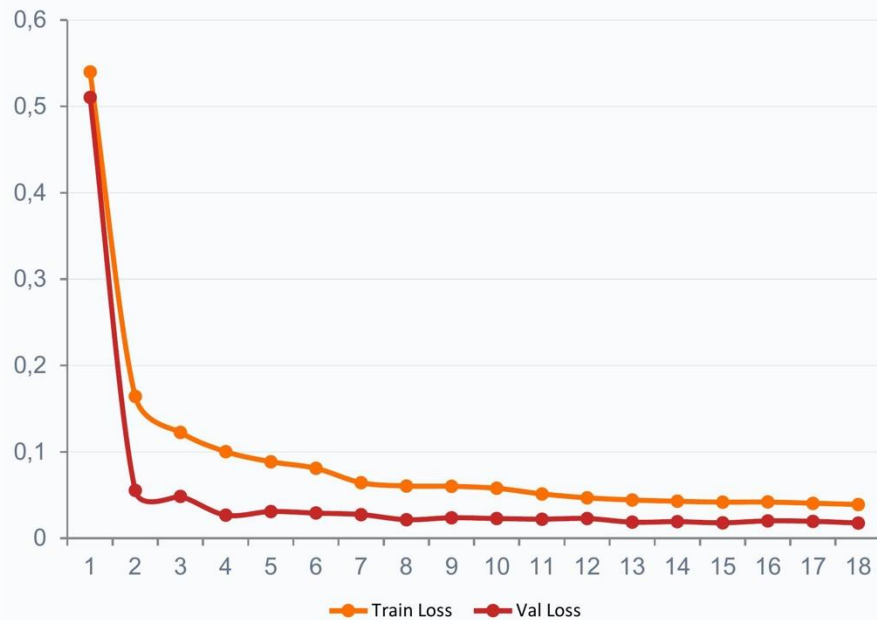
# Training Progress : Accuracy & Loss Curves

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Accuracy per Epoch



Loss per Epoch

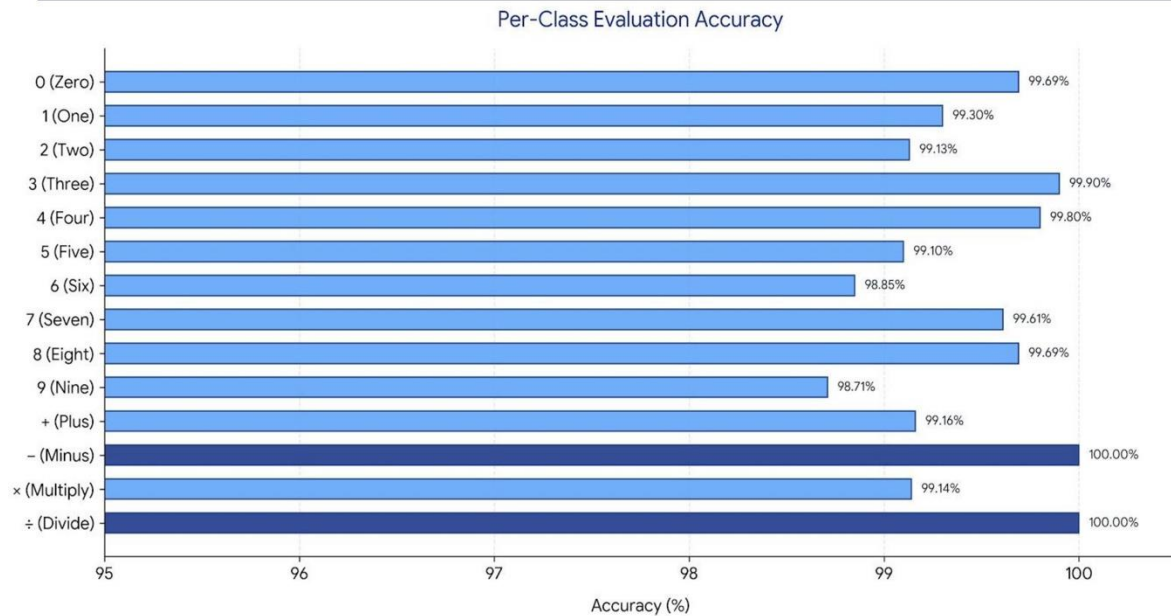


Fast convergence: val acc jumped from 85.7% → 98.4% in just 2 epochs

Best epoch: 13 (val acc 99.39%, val loss 0.0186)

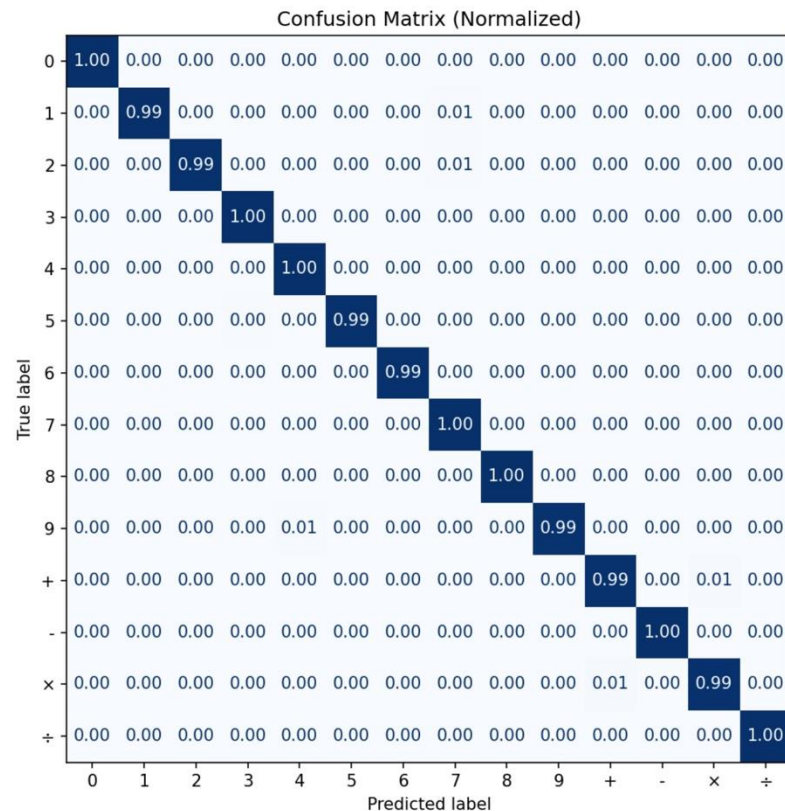
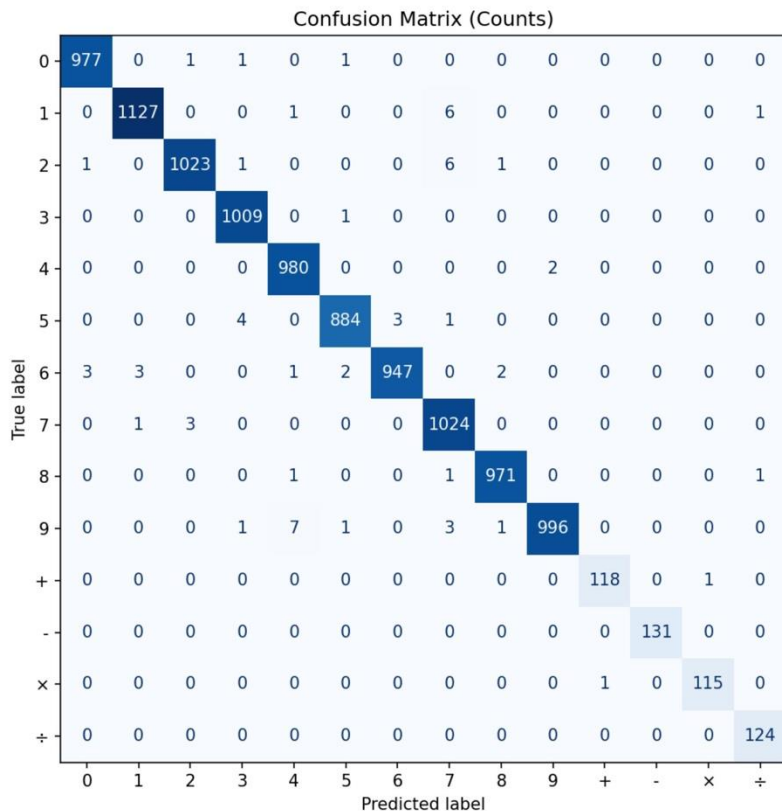
Early stopping triggered at epoch 18, no overfitting observed

ReduceLROnPlateau halved LR at epochs 6 / 10 / 12 / 17



# Confusion Matrix

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## Top confusion pairs (test set, 59 total errors)

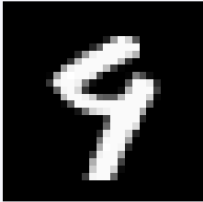
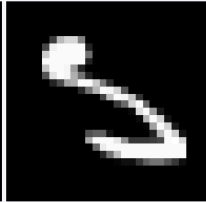
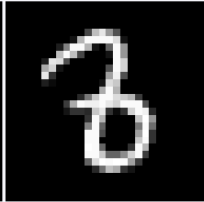
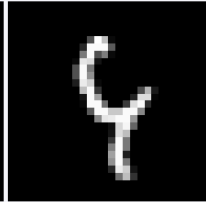
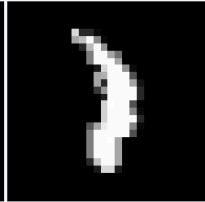
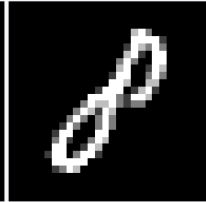
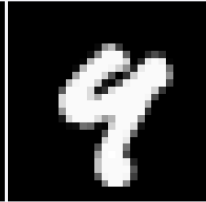
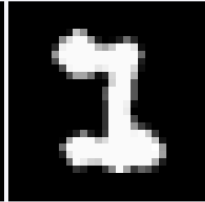
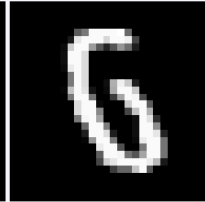
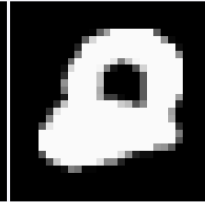
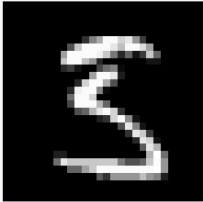
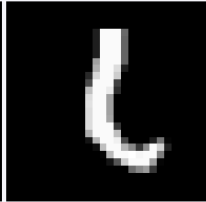
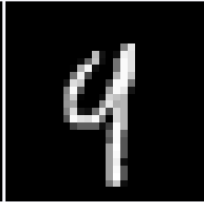
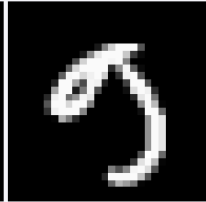
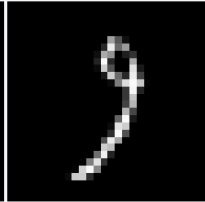
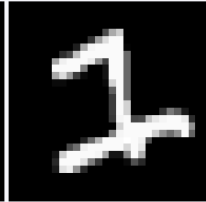
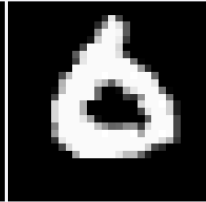
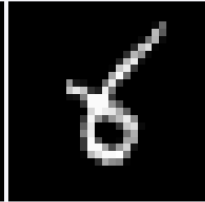
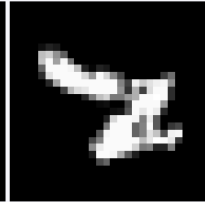
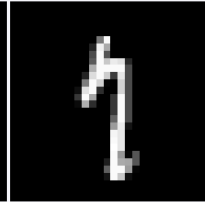
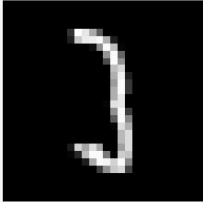
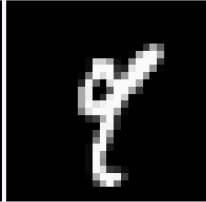
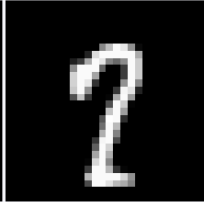
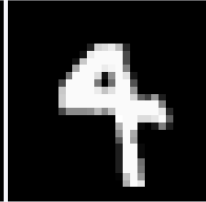
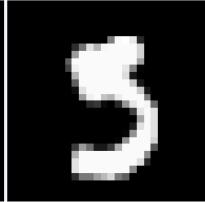
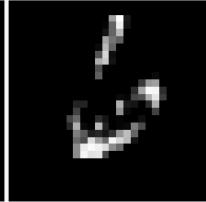
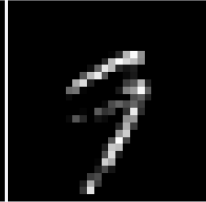
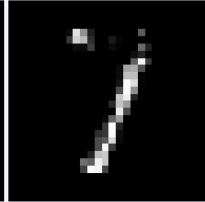
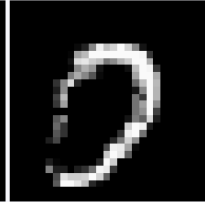
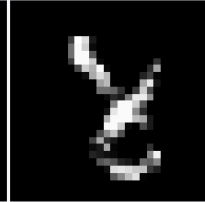
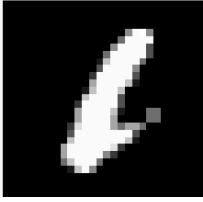
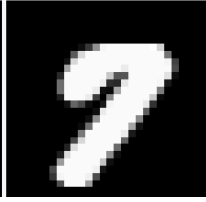
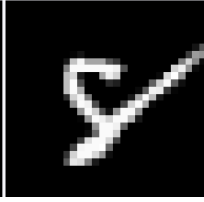
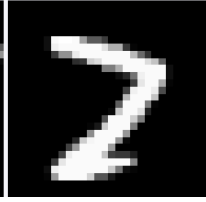
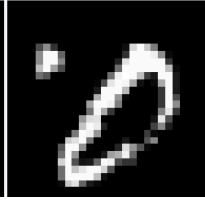
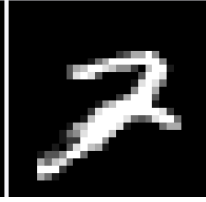
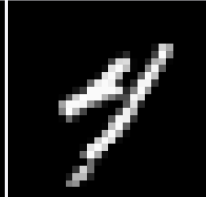
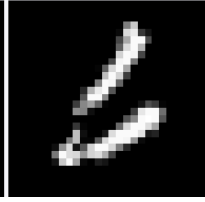
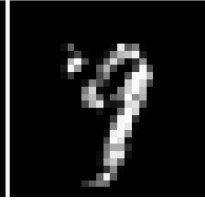
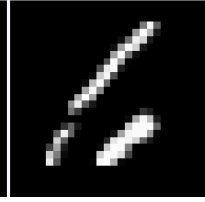
| True | Predicted | Count | Why?                           |
|------|-----------|-------|--------------------------------|
| 4    | 9         | 5     | Open loop vs. stem             |
| 5    | 3         | 5     | Similar curvature              |
| 9    | 4         | 4     | Open loop vs. stem             |
| 7    | 2         | 4     | Diagonal bar confusion         |
| ×    | 7         | 3     | Cross $\approx$ slanted stroke |

## Key observations

- 9 is the weakest class: 98.7% accuracy
- – and  $\div$ : 100% accuracy on the test set
- Most errors are **digit-to-digit**, not cross-type
- Worst pairs are visually near-identical shapes

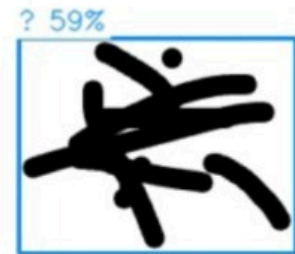
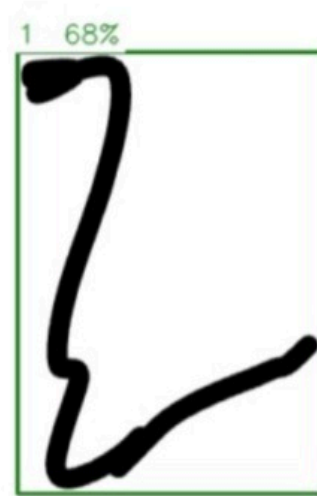
With only 1 048 operator samples, × is most sensitive to style variation — being confused as a slanted 7.

# Failure Cases — All 59 Misclassified Symbols

|                                                                                    |                                                                                     |                                                                                     |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                       |
|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| ✓9 → X4<br>70%                                                                     | ✓5 → X3<br>68%                                                                      | ✓8 → X3<br>54%                                                                      | ✓9 → X4<br>97%                                                                      | ✓1 → X2<br>53%                                                                       | ✓8 → X0<br>55%                                                                        | ✓4 → X9<br>97%                                                                        | ✓1 → X3<br>47%                                                                        | ✓6 → X0<br>58%                                                                        | ✓2 → X0<br>67%                                                                        |
|    |    |    |    |    |    |    |    |    |    |
| ✓5 → X3<br>69%                                                                     | ✓6 → X1<br>90%                                                                      | ✓4 → X9<br>91%                                                                      | ✓9 → X5<br>61%                                                                      | ✓9 → X7<br>67%                                                                       | ✓1 → X2<br>51%                                                                        | ✓6 → X0<br>87%                                                                        | ✓6 → X8<br>86%                                                                        | ✓2 → X7<br>86%                                                                        | ✓1 → X7<br>81%                                                                        |
|    |    |    |    |    |    |    |    |    |    |
| ✓3 → X2<br>76%                                                                     | ✓9 → X4<br>45%                                                                      | ✓7 → X2<br>72%                                                                      | ✓4 → X9<br>41%                                                                      | ✓5 → X3<br>50%                                                                       | ✓6 → X÷<br>59%                                                                        | ✓9 → X7<br>60%                                                                        | ✓7 → X1<br>99%                                                                        | ✓0 → X7<br>64%                                                                        | ✓8 → X2<br>45%                                                                        |
|    |    |    |    |    |    |    |    |    |    |
| ✓1 → X6<br>71%                                                                     | ✓7 → X2<br>54%                                                                      | ✓8 → X4<br>98%                                                                      | ✓7 → X2<br>87%                                                                      | ✓0 → X2<br>71%                                                                       | ✓2 → X7<br>91%                                                                        | ✓4 → X7<br>88%                                                                        | ✓6 → X1<br>91%                                                                        | ✓9 → X7<br>48%                                                                        | ✓6 → X1<br>99%                                                                        |
|  |  |  |  |  |  |  |  |  |  |

## Solver needs single characters

- probability vector gets converted to char
- confidence > 60% → decision
- otherwise return “?”



## Solver

- chars merged to string expression
- expression solved with AST evaluator

**99.44%** Test accuracy

**14** Unified classes

**178K** Parameters

**< 3 min** Training time

**What we built** — End-to-end math solver running in the browser.

- One unified CNN — digits **and** operators
- Safe AST evaluator (no exec / eval)
- OpenCV segmentation with proximity merging

Adding a new symbol class = more training data only, no new code.

**Future work** — Next steps to extend the project.

- Multi-digit numbers as a single unit
- Fractions and parentheses
- More classes: =, decimal point, variables
- Larger operator dataset → reduce failure cases
- Mobile / touch input optimization

# Thank You

Questions?

Matti H. V. Lehmann

David A. Spickenheuer

Jonas Schenke

Philip Nguyen

Staniya Thomas

Abhirup Sain